

THE WEST AFRICAN EXAMINATIONS COUNCIL

BASIC EDUCATION CERTIFICATE EXAMINATION -- JUNE 2008

INTEGRATED SCIENCE -- PAPER 1

Duration: 1 Hour | Level: JHS 3 / BECE | Code: SCI 01/02 - P1

INSTRUCTIONS TO CANDIDATES

Answer all 40 questions on your objective answer sheet. Each question is followed by four options lettered A to D. Choose the correct option and mark it with 2B pencil.

PART I: OBJECTIVE QUESTIONS

1. Which of the following cellular structures is present in plant cells but absent in animal cells?

- A. Cell membrane
- B. Cytoplasm
- C. Cellulose cell wall
- D. Mitochondria

2. The process by which green plants manufacture carbohydrates using sunlight is called...

- A. Transpiration
- B. Respiration
- C. Photosynthesis
- D. Fermentation

3. Which of the following is an example of a chemical change?

- A. Melting of ice
- B. Rusting of iron
- C. Dissolving sugar in water
- D. Evaporation of alcohol

4. The SI unit for measuring electrical resistance is...

- A. Ampere
- B. Volt
- C. Ohm
- D. Watt

5. Which of the following gas components makes up approximately 78% of atmospheric air?

- A. Oxygen
- B. Carbon dioxide
- C. Nitrogen
- D. Argon

6. A substance with a pH value of 3.0 is classified as...

- A. Strongly acidic
- B. Weakly acidic
- C. Neutral
- D. Basic

7. Which of the following organisms causes malaria in humans?

- A. Female Anopheles mosquito
- B. Plasmodium parasite
- C. Tsetse fly
- D. Housefly

8. The bending of a ray of light as it passes from air into water is known as...

- A. Reflection
- B. Refraction
- C. Dispersion
- D. Diffraction

9. Which part of the human alimentary canal is primarily responsible for the absorption of digested nutrients?
 - A. Stomach
 - B. Duodenum
 - C. Ileum (Small intestine)
 - D. Colon
10. An example of a first class lever is...
 - A. Wheelbarrow
 - B. Pair of scissors
 - C. Nutcracker
 - D. Fishing rod
11. The process of removing metabolic waste products from the body of a living organism is called...
 - A. Egestion
 - B. Excretion
 - C. Secretion
 - D. Defecation
12. Which of the following soil types has the highest water retention capacity?
 - A. Sandy soil
 - B. Loamy soil
 - C. Clay soil
 - D. Gravel
13. What is the chemical formula for ordinary table salt?
 - A. NaOH
 - B. NaCl
 - C. CaCO₃
 - D. HCl
14. The transfer of heat through solids from a region of higher temperature to lower temperature without movement of the material is...
 - A. Conduction
 - B. Convection
 - C. Radiation
 - D. Evaporation
15. Which vitamin is synthesized in the human skin upon exposure to morning sunlight?
 - A. Vitamin A
 - B. Vitamin B
 - C. Vitamin C
 - D. Vitamin D
16. The male reproductive organ of a flowering plant is known as the...
 - A. Carpel
 - B. Pistil
 - C. Stamen
 - D. Petal
17. Which of the following agricultural practices directly prevents soil erosion on steep hillsides?
 - A. Terracing
 - B. Slash and burn
 - C. Monoculture
 - D. Overgrazing
18. The density of a substance is defined as...
 - A. Mass x Volume
 - B. Mass / Volume
 - C. Weight / Volume
 - D. Volume / Mass
19. Which blood cell type is responsible for defending the human body against infectious pathogens?
 - A. Red blood cells
 - B. White blood cells
 - C. Platelets
 - D. Plasma

20. The chemical symbol for Potassium is...

- A. P
- B. Pt
- C. K
- D. Po

21. An echo is produced due to the...

- A. Refraction of sound
- B. Reflection of sound
- C. Absorption of sound
- D. Diffraction of sound

22. Which of the following disease conditions is caused by a deficiency of iodine in the human diet?

- A. Rickets
- B. Scurvy
- C. Goitre
- D. Beriberi

23. The process by which water vapour cools and turns into liquid droplets is called...

- A. Sublimation
- B. Evaporation
- C. Condensation
- D. Freezing

24. Which gas is evolved when dilute hydrochloric acid reacts with calcium carbonate?

- A. Oxygen
- B. Hydrogen
- C. Carbon dioxide
- D. Chlorine

25. What type of energy transformation occurs in a microphone?

- A. Electrical to sound
- B. Sound to electrical
- C. Mechanical to thermal
- D. Chemical to light

26. A safety device in an electrical circuit that melts to break the circuit when excessive current flows is a...

- A. Switch
- B. Fuse
- C. Resistor
- D. Transformer

27. Which planet is closest to the Sun in our Solar System?

- A. Venus
- B. Mercury
- C. Earth
- D. Mars

28. The loss of water in the form of water vapour from the aerial parts of a plant is termed...

- A. Guttation
- B. Transpiration
- C. Osmosis
- D. Diffusion

29. Which of the following instruments is used to measure atmospheric pressure?

- A. Thermometer
- B. Barometer
- C. Hydrometer
- D. Anemometer

30. What is the chemical name of rust?

- A. Hydrated iron(III) oxide
- B. Iron(II) chloride
- C. Iron(II) sulphate
- D. Iron sulphide

31. Which organ in the human body filters urea from the bloodstream to form urine?
A. Liver
B. Kidney
C. Lungs
D. Heart
32. The process of heating milk to kill disease-causing microorganisms is called...
A. Sterilization
B. Pasteurization
C. Fermentation
D. Refrigeration
33. In magnetism, like poles...
A. Attract
B. Repel
C. Neutralize
D. Do not interact
34. Which of the following is a non-renewable source of energy?
A. Solar energy
B. Crude oil
C. Wind energy
D. Biomass
35. The component of a flower that attracts insect pollinators is the...
A. Sepal
B. Petal
C. Receptacle
D. Filament
36. The boiling point of pure water at standard atmospheric pressure is...
A. 0 deg C
B. 50 deg C
C. 100 deg C
D. 212 deg C
37. Which gas is essential for human aerobic cellular respiration?
A. Carbon monoxide
B. Nitrogen
C. Oxygen
D. Helium
38. A mixture of iron filings and sulphur powder can be separated physically by using a...
A. Filter paper
B. Magnet
C. Separating funnel
D. Condenser
39. The female gamete in humans is produced in the...
A. Uterus
B. Ovary
C. Fallopian tube
D. Vagina
40. Which of the following is a biodegradable waste material?
A. Polythene bag
B. Glass bottle
C. Banana peel
D. Aluminium can

OFFICIAL WAEC ANSWER KEY & SCORING GUIDE

For candidate self-assessment and teacher evaluation:

1. [C] 2. [C] 3. [B] 4. [C] 5. [C]
6. [A] 7. [B] 8. [B] 9. [C] 10. [B]
11. [B] 12. [C] 13. [B] 14. [A] 15. [D]

16. [C] 17. [A] 18. [B] 19. [B] 20. [C]
21. [B] 22. [C] 23. [C] 24. [C] 25. [B]
26. [B] 27. [B] 28. [B] 29. [B] 30. [A]
31. [B] 32. [B] 33. [B] 34. [B] 35. [B]
36. [C] 37. [C] 38. [B] 39. [B] 40. [C]